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Power Price Forecasting in Japan: Methods, Models, and Insights

Introduction

Japan's electricity market has undergone major changes in recent years, including full deregulation (around 2016) and rapid growth of renewable energy. These shifts have led to increased price volatility and even instances of extremely low or zero prices during high renewable output periods [1]. Accurate day-ahead electricity price forecasting (DAEPF) in Japan is therefore crucial for utilities, traders, and renewable producers to manage risks and optimize bidding strategies [2] [3]. This report surveys power price forecasting methods used in academic research and industry literature, highlighting recent (last ~5 years) studies in Japan and drawing comparisons with broader international approaches where relevant. We cover the spectrum of forecasting models (from traditional statistical methods to modern machine learning), the input features and feature engineering techniques they employ, common performance metrics, and validation practices (such as rolling forecast evaluation) used to ensure robust predictions.

Forecasting Methods and Models in Literature

Traditional Statistical Approaches

Early and foundational work on electricity price forecasting often relied on **time-series statistical models**. The most widely used are Autoregressive Integrated Moving Average (**ARIMA**) and related formulations. ARIMA models (and simpler ARMA variants) have been applied to day-ahead price prediction in many markets [4]. For example, researchers successfully fit ARIMA models to Spain and California price data [5], and an ARIMA(4,1,2) was found best for UK prices in one study (performing better than persistence and even a basic neural network) [6]. Seasonal ARIMA with exogenous inputs (SARIMAX) has also been used to capture weekly/annual seasonality in prices [7]. These linear models provide a good baseline but struggle with the pronounced non-linearities and abrupt spikes in power prices [8].

To handle **heteroskedasticity and spikes**, researchers introduced volatility and regime-switching models. **GARCH (Generalized Autoregressive Conditional Heteroskedasticity)** models (including Markov-switching GARCH) have been used to model time-varying price volatility [9]. A study on Nordic prices combined a Markov-switching GARCH to capture regime-dependent variance, confirming that price volatility often follows different regimes (e.g. normal vs. spike periods) [10]. Similarly, regime-switching stochastic models with time-varying parameters have been proposed to mimic the jumpy behavior of spot prices, using indicators like reserve margin and system load to switch between "normal" and high-price regimes [11]. These statistical models improved representation of **price spikes**, which are rare events where prices shoot far above normal levels. Some works treat spike occurrences as separate processes – for instance, **Poisson or count-process models** have been used to forecast the probability of extreme price events [12] [13]. An example in Japan is the **Hawkes process** approach: Adline and Ikeda (2023) modelled JEPX price spikes (prices >25 JPY/kWh) as a self-exciting point process, where one spike increases the likelihood of subsequent spikes in the short term [14] [15]. Their modified Hawkes model outperformed other methods

for multi-day spike prediction, evaluated by metrics like MAE of spike probability and Matthews correlation coefficient [16]. In summary, classical models (ARIMA/SARIMA, GARCH, regime-switching) set the foundation and are still used as benchmarks, but they often require enhancements to handle the **non-linear, non-stationary** nature of modern electricity prices in Japan.

Machine Learning and AI Approaches

In the past decade, **machine learning (ML)** techniques have gained prominence in electricity price forecasting, often outperforming the above statistical models [17]. Early ML approaches included **artificial neural networks (ANN)** and **support vector machines (SVM)**. For example, an ANN was used with fundamental inputs to predict next-day prices [18], though one comparative study noted a tuned ARIMA could still beat a "typical" ANN in some cases [6]. Support vector regression (SVR), especially with kernel functions, has also been applied; Fang *et al.* (2024) tested ANN vs. SVR models for JEPX prices and found that a simple "narrow" ANN and a linear-kernel SVR actually performed best among various configurations [19] [20]. Their study underscores that when the correct variables are used (in their case, **net demand = demand minus PV output**), even relatively simple ML models can yield strong accuracy, especially if retrained continuously (they used a "continuous loop" training method akin to rolling updates) [21].

Beyond ANN/SVM, tree-based ensemble methods like **Random Forests** and **Gradient Boosting (e.g. XGBoost)** have become popular for price forecasting. These models handle non-linearity and interactions well and offer interpretability of feature importance. In one benchmark, **XGBoost, LSTM, and deep neural networks outperformed more complex hybrid deep models** on price data [22]. (Notably, that study also observed that overly complex CNN or RNN architectures and elaborate hybrid schemes did not always improve accuracy, and simple averaging ensembles could rival their performance [22].) Such results align with Lago *et al.* (2018), who compared 27 algorithms and found that on average **deep learning models outperform classical models**, but that hybrids did not consistently beat single models [17].

Deep learning approaches are now state-of-the-art in many academic studies. **Recurrent neural networks (RNNs)** tailored for sequences have been extensively used – e.g. a gated recurrent unit (GRU) network was proposed for price forecasting [23]. The most popular are **LSTM (Long Short-Term Memory)** networks, which capture long-range temporal dependencies in price series. LSTMs have shown strong performance in electricity markets worldwide and in Japan [24]. Many works augment LSTM models with additional features or hybrid structures. A notable approach is combining LSTM with a **Convolutional Neural Network (CNN)**. The CNN can automatically extract local patterns (e.g. recent trends) from input sequences, feeding into the LSTM which models time dependencies. This hybrid **CNN–LSTM** model has proven very effective: Cordoni *et al.* compared CNN–LSTM against standalone LSTM, plain CNN, multilayer perceptron, random forest, etc., and found CNN–LSTM achieved the highest accuracy for day-ahead price forecasting [25]. In that study, input features included past prices and loads plus calendar info, though no weather data [26]. More recently, Wang *et al.* (2024) applied a CNN–LSTM ensemble in Japan's Kyushu area and demonstrated clear gains over a single LSTM, both in accuracy and training speed [27]. By using CNN to extract features, their training time was shorter than LSTM alone, yet accuracy improved [24]. In fact, **ensemble learning** was used to further boost robustness: they trained multiple CNN–LSTM models with different initializations and averaged the predictions, significantly reducing variance and improving overall MAE/RMSE [28] [29].

The frontier of ML in price forecasting is moving toward advanced architectures. For example, **attention-based models and Transformers** are being explored internationally. A 2024 study tested the **Temporal Fusion Transformer (TFT)** and **N-BEATS** (Neural Basis Expansion) models on simulated future price

scenarios, finding they delivered high accuracy without needing scenario-specific retraining [30]. These sophisticated models can incorporate multiple covariates and output probabilistic forecasts, though as of now they have not seen wide application in Japan-focused studies. Nonetheless, their success in Europe and elsewhere suggests they may soon be adopted for JEPX price forecasting as data availability grows.

Finally, many researchers experiment with **hybrid models** that combine methodologies or data preprocessing techniques. For instance, an approach combined ARFIMA (a fractional ARIMA) with an SVM to leverage both long-memory and machine learning components [31]. Hybrid **denoising methods** are also common: applying wavelet transforms or empirical mode decomposition (EMD) to price series before forecasting. A hybrid model that applied maximal overlap wavelet transform (MODWT) filtering plus EMD, followed by an LSTM network, outperformed a standard LSTM in one case [32]. Another hybrid angle addresses **outliers**: Wang *et al.* (2020) developed an outlier-robust forecasting model combining an Extreme Learning Machine (ELM) with other algorithms, which proved effective on markets with frequent spikes (they demonstrated this on Australia's and Singapore's markets) [33]. The inclusion of a robust component helped handle the **numerous outliers and nonlinearities** in price data [33]. In summary, machine learning approaches – especially LSTM-based deep networks and ensemble tree models – now dominate recent literature on Japan's electricity price forecasting, generally providing better accuracy than classic models. However, researchers have also noted that **model simplicity vs. complexity must be balanced**: sometimes a well-tuned simpler model (e.g. linear SVR or shallow ANN) with the right features can rival more complex networks, especially when data is limited or model retraining is frequent [19] [20].

Table 1 below highlights a few representative studies of price forecasting in Japan, illustrating the range of models and findings:

Study & Year	Methods Compared	Best/Notable Model	Key Features Used	Performance Metrics	Noteworthy Findings
Wang *et al.* (2024) [27] [34]	CNN–LSTM vs. LSTM (ensemble approach)	CNN–LSTM ensemble model	Area price, system price, actual generation (PV), weather forecasts, calendar (day type), **rolling window stats** of price [35]	MAE ≈ 1.94 JPY/kWh, RMSE ≈ 2.63 JPY/kWh on test [34]	CNN–LSTM outperforms single LSTM in accuracy and ~2× faster training [24]. Using **all features** yielded best accuracy. Employed daily re-training (rolling) for day-by-day forecasts [36], and ensemble averaging of 30 runs to stabilize predictions [37] [29].

Study & Year	Methods Compared	Best/Notable Model	Key Features Used	Performance Metrics	Noteworthy Findings
Fang *et al.* (2024) [38] [21]	ANN vs. SVR (different network widths, kernels, and training loops)	**"Narrow" ANN** (shallow NN) & **Linear SVR**	Net demand (total load minus PV output) as a primary feature, plus total demand and PV generation levels	(Reported relative comparison: ANN/SVR outperformed others; exact MAE not given in abstract)	Emphasizes **net demand** as a crucial feature reflecting solar penetration [38]. Found that simpler models (a small ANN and linear SVR) **with continuous retraining** (rolling "continuous loop" training) **beat more complex settings** [21] – highlighting the value of frequent model update and appropriate feature choice in Japan's solar-influenced market.

Study & Year	Methods Compared	Best/Notable Model	Key Features Used	Performance Metrics	Noteworthy Findings
Kaneko & Inoue (2023) [39] [40]	Sparse linear modeling (Lasso-based regression) for factor analysis	n/a (focus on interpretability, not a specific predictive model benchmark)	Past prices, weather factors (temperature, etc.), calendar indicators (weekday/ holiday), and proxies for consumer lifestyle/ behavior	(Performance not the focus; results evaluated in terms of representation error and interpretability)	Introduced a **sparse modeling** approach to identify key drivers of JEPX prices [41]. The method automatically selected significant variables; **calendar variables (day-of-week, holidays)** emerged as most influential, even more than weather or other factors [40]. This suggests seasonality/ lifestyle patterns strongly affect price dynamics in Japan.

Study & Year	Methods Compared	Best/Notable Model	Key Features Used	Performance Metrics	Noteworthy Findings
Nakayama *et al.* (2023) [42]	Machine learning model integrating **GIS-based features**	ML model (type not specified, uses derived GIS inputs)	Geospatial distribution of supply/ demand: e.g. regional solar capacity, location of generation and demand centers, meteorological data mapped by location	Achieved lower forecast error than a baseline model without GIS features (exact error improvement not quantified in abstract)	Pioneered the use of **geographic information system (GIS)** data in price forecasting. By deriving explanatory variables linked to location (e.g. solar irradiance over areas, regional demand-supply balances), they improved day-ahead price forecast accuracy beyond existing methods [42]. Illustrates the value of spatial features (especially to capture solar generation variability across Japan).

Features Used in Price Forecasting Models

The accuracy of any forecasting model depends heavily on the input features. In Japan-focused studies, researchers incorporate a wide range of **explanatory variables** to capture the drivers of electricity price fluctuations:

- **Historical Prices:** Almost all models include past price values. Lagged prices (from previous hours/ days) provide the baseline trend and mean-reversion patterns. In advanced models, not only individual lags but also **rolling window statistics** are used – e.g. Wang *et al.* calculated the 3-day

rolling **min, max, mean, and std deviation** of past prices as additional features [43]. These capture recent volatility and range, which help predict next-day price levels.

- **Demand (Load):** System demand or regional load is a fundamental driver of price. Higher demand tends to raise prices, all else equal. Many studies include total load as an input [26]. Some use related measures like **net demand** (demand minus renewable output) – Fang *et al.* showed that using net demand was crucial in reflecting Japan's solar-driven midday price dips [38]. Demand is often available by area (e.g. Tokyo area load), which can be matched to area prices for regional forecasting.

- **Generation & Renewable Output:** With Japan's increasing renewable penetration, features describing generation supply have become important. This includes **renewable generation forecasts** (especially solar PV and wind output) and conventional generation status. For example, the Kyushu CNN-LSTM study incorporated the area's actual power generation and system-wide generation mix implicitly through the system price [35]. Solar output in particular has a strong influence on daytime prices (large PV output can push prices down). Thus, some models take *forecasted irradiance* or expected PV generation as inputs [35]. On the flip side, availability of thermal plants or fuel-based generation might be indirectly reflected via fuel prices or reserve margin (see below). In general, supply-side factors aim to inform the model about **capacity margins** – tight supply (low reserve) leads to spikes, oversupply (e.g. sunny low-demand days) leads to low prices.

- **Weather Variables:** Weather affects both demand and renewable supply, so it's commonly used. Temperature is a key driver of demand (hot summers and cold winters cause demand peaks from AC/heating). Solar irradiance and wind speed affect renewable generation. Many models include temperature forecasts and sometimes humidity or wind data for the regions of interest. Studies have *validated that weather data significantly improves day-ahead price forecasts* [44]. For instance, including temperature and solar radiation can help anticipate unusual price drops due to mild weather or zero-price periods during high solar production [1] [35]. (An example outside Japan: in the Turkish market, fluctuations in renewable output were fundamental to explaining price spikes [45] – the principle holds in Japan with its solar output swings.) Weather feeds both demand-side (via temperature-sensitive load) and supply-side (via renewables), making it a high-impact feature category.

- **Fuel and Cost Factors:** Fuel prices (oil, LNG, coal) and related costs influence the marginal cost of thermal generators, thus impacting wholesale prices. Japan relies on imported fuels, so fuel price volatility can translate to electricity price trends. Some studies mention fuel cost as an important factor [46]. For example, if LNG prices surge, gas-fired generation offers become expensive, raising market clearing prices. While short-term (day-ahead) forecasts may not always include fuel prices (which move more slowly than daily prices), **month-ahead or scenario analyses** do consider them. In at least one study on the Korean market (which has similarities in fuel import dependence), monthly coal and gas prices were included for day-ahead predictions [47]. In Japan, fuel price could be more relevant for medium-term forecasting or stress-test scenarios rather than day-to-day prediction, but it remains an underlying driver of price levels.

- **Calendar and Seasonality: Calendar effects** are consistently found to be critical features. These include day-of-week (weekday vs weekend), hour-of-day, month or season, and holidays. Electricity demand in Japan exhibits strong weekly cycles (lower demand and price on weekends) and seasonal patterns (high summer and winter demand). Kaneko & Inoue's sparse model analysis found that

calendar indicators were among the most important variables, even more influential than weather or past price in their model [40]. Many forecasting models therefore add dummy variables or cyclical encodings for weekdays, weekends, holidays, and perhaps seasonal indices. Some also treat **hourly cycles** using 24-hour sine/cosine features to capture intraday patterns (peak price typically in evening, trough late night, etc.). Including calendar variables improves baseline accuracy by accounting for regular predictable fluctuations (e.g. Monday morning demand surge, Golden Week holiday slump). In the CNN–LSTM study, for instance, adding "calendar forecasts" (day type features) contributed to accuracy gains [35].

- **Market and Operational Data:** Other features can include **market-specific signals** such as *reserve margin*, *interconnector flows/constraints*, or *bidding behavior indicators*. For example, a study of Italy's market found that the shape of the supply curve (from auction data) and available reserve were determinants of price spikes [11]. In Japan, granular bidding data is not always accessible publicly, but proxies exist: **system price vs area price** differences can indicate congestion; the system-wide demand-supply balance can be derived from total generation vs load. The **JEPX system price** (uniform price with no grid constraints) was used alongside area prices by Wang *et al.* [35], effectively providing a market-wide price signal as a feature for regional price forecasting. Additionally, grid constraints (like interregional transmission limits) occasionally drive price separation between areas (e.g. Hokkaido often has the highest area price due to limited interconnection [48]). While these are harder to include directly, some models implicitly capture them by treating each area separately or including **regional dummy variables**.

- **Unconventional data sources:** A few innovative studies have incorporated non-traditional data to improve forecasts. One example is **web search trends**: Gotoh (2024) examined how Google search keyword volumes (e.g. related to weather or energy) correlate with JEPX price movements [49]. They found a moderate positive correlation between spikes in certain keyword searches and subsequent price increases, suggesting that public interest or concern (as reflected in searches) could serve as an early indicator of demand surges or market tightness [49]. While not yet common, such *social data* could enrich forecasting models. Another novel angle is the aforementioned **GIS-based features**: by mapping the spatial distribution of solar farms and demand centers, Nakayama *et al.* derived features that encapsulate the **geographical supply-demand balance**, which improved price prediction [42]. These examples show how researchers are reaching beyond typical time-series data, attempting to capture the more complex drivers of price (from human behavior to location-dependent effects).

In summary, effective price forecasting models in Japan tend to use a **wide array of features**: historical prices and loads, weather and renewable output data, calendar/seasonal indicators, and occasionally external signals like fuel costs or novel proxies. Feature importance analyses confirm that **demand (especially net-demand), renewable generation, and calendar effects** are among the top drivers for the Japanese market's price dynamics [38] [40]. Incorporating all these feature types can significantly enhance accuracy – for instance, in a comprehensive model, adding *all proposed features* (price history, system price, generation, weather, calendar, etc.) yielded the highest forecasting accuracy, as evidenced by the CNN–LSTM model's results [50].

Feature Engineering Techniques

To extract maximal value from the raw features, various **feature engineering** techniques are employed in the literature. These techniques transform, select, or augment the input data to help models learn the underlying patterns more effectively:

- **Rolling Window Statistics:** As mentioned, computing aggregate statistics over a recent rolling window is a common technique for time series. By taking, say, the last 72 hours (3 days) of prices and calculating features like minimum, maximum, mean, and standard deviation, models gain information about the recent price range and volatility [43]. This helps in a volatile market by providing context (e.g. whether the current price is unusually high or low relative to the past few days). Wang *et al.* (2024) explicitly included a 3-day rolling window for price, noting it was *"crucial for capturing temporal patterns and trends"* in the data [43].

- **Log or Power Transformations:** Price distributions are often skewed with occasional extreme spikes. Transforming the target variable can stabilize variance and improve model fitting. A simple yet effective method is taking the **natural logarithm** of prices. In the Kyushu price forecast study, applying a log transform to the target price improved prediction accuracy noticeably [51]. The log compresses the scale of spikes (e.g. 100 JPY becomes ~4.6 in log terms), making it easier for models to learn relationships without being dominated by rare extreme values. After forecasting in log-scale, the predictions are exponentiated back to normal prices. Other transformations like square root or Box-Cox could similarly be used depending on data characteristics, though log is most common in recent papers [51].

- **Decomposition and Denoising:** To handle complex seasonality and noise, **time-series decomposition** methods are used in some advanced pipelines. For example, one might decompose the price or demand series into trend, seasonal, and residual components (using methods like STL or MSTL – Seasonal and Trend decomposition using Loess) [52] [53]. By doing so, a model (like LSTM or XGBoost) can be trained on the *seasonal* component separately or on a de-noised series. In a Korean price forecasting study, several decomposition techniques (Fourier transforms, wavelets, Variational Mode Decomposition, STL/MSTL) were tested, and incorporating them helped reduce overfitting and improved accuracy for certain models [54] [55]. A specific example is using **wavelet transforms** to remove high-frequency noise: a hybrid model with MODWT wavelet denoising + EMD (Empirical Mode Decomp.) feeding into LSTM achieved better forecasts than directly using raw price data [32]. Essentially, these techniques **pre-filter the data**, isolating predictable patterns (like daily cycles or long-term trend) from irregular fluctuations, thus aiding the learning algorithm.

- **Outlier Detection and Treatment:** Electricity prices can exhibit outliers (spikes or occasional negative/zero prices). These outliers can distort model training if not handled. One strategy is to identify and replace outlier values in the training data (e.g. capping extreme values or interpolating). Another is to use models robust to outliers. Wang *et al.* (2020) introduced an **outlier-robust ELM** in their hybrid model which specifically aimed to reduce the influence of abnormal price points [33]. Additionally, some two-stage approaches first **predict the occurrence of spikes** (e.g. via classification or threshold models) and then model the price magnitude. The "policy-versus-policy" two-stage strategy in one Japanese study is an example: it first handled predicting zero-price events and then adjusted the price forecasting accordingly [56]. By separating the spike/no-spike decision,

the model could use different regimes for normal vs. extreme periods. Overall, explicitly accounting for outliers leads to more reliable forecasts especially in volatile markets.

- **Feature Selection & Dimensionality Reduction:** With many potential inputs (dozens of weather features, lags, etc.), some studies apply techniques to **select the most relevant features**. Kaneko & Inoue (2023) used a *sparse modeling* (Lasso regression) approach to automatically pick important variables based on non-zero coefficients [41]. This not only improves interpretability but can also enhance performance by removing redundant inputs. Similarly, tree-based models can rank feature importance, and this can guide a manual selection of top predictors. In one approach, Random Forest feature importance was computed prior to modeling to filter inputs, ensuring only significantly predictive variables were fed to the final model [57]. Another approach is **principal component analysis (PCA)** or similar dimensionality reduction on correlated inputs (though PCA is less common in recent deep learning pipelines, where the network can handle many inputs). The key idea is to prevent overfitting and reduce noise by focusing the model on a subset of features that carry most information about price movements [58].

- **Encoding Cyclical Features:** As mentioned, representing cyclical time features (hour of day, day of week, month of year) in a model-friendly way is a form of feature engineering. Rather than using an integer 1–24 for hour (which misrepresents the cyclical nature), many use sine/cosine transforms to encode hour-of-day or day-of-week. This ensures that 23:00 and 00:00 are recognized as adjacent in the cycle, for example. While not always explicitly detailed in papers, this practice is widespread in industry implementations of load and price forecasting. It complements including dummy variables or one-hot encodings for daily/weekly cycles.

- **Ensemble and Meta-learner Techniques:** Although more of a modeling strategy than feature engineering per se, combining multiple models can be viewed as creating new "features" (the predictions from each model) that are then averaged or input to a meta-model. The CNN–LSTM ensemble is one example where multiple networks' outputs were averaged [37]. In other cases, researchers might take a **blend of models** (say, one ARIMA and one XGBoost) and average their forecasts, effectively using the model type as a feature. There are even studies where a secondary model (like a small neural network) learns to weight the different model outputs – sometimes called **stacking** or **expert selection** [59]. For instance, Neupane *et al.* (2017) developed an ensemble with expert selection for EPF [60]. Such techniques acknowledge that different models have different strengths under various conditions, so combining them can yield a more robust prediction.

In practice, applying these feature engineering techniques has been shown to improve forecast performance significantly. For example, after fully integrating *all* their engineered features (log-transformed prices, rolling stats, calendar, weather, etc.), the CNN–LSTM ensemble in Kyushu achieved its highest accuracy (MAE ~1.936 JPY, RMSE ~2.630 JPY) [50]. Likewise, using feature selection in the sparse model allowed Kaneko *et al.* to pinpoint that calendar effects dominated, an insight that can simplify future models [40]. The careful crafting of input features – through statistical transforms, decomposition, or selection – is thus a critical step in both academic studies and industry practice for electricity price forecasting.

Performance Evaluation Metrics

Researchers evaluate forecasting accuracy using standard error metrics common to time series forecasting. In the electricity price domain (including Japanese market studies), the most widely reported **error terms** or metrics are:

- **Mean Absolute Error (MAE):** the average of absolute differences between predicted and actual prices. MAE is easy to interpret (in the same units as price). For instance, the CNN–LSTM model achieved a MAE of ~1.936 JPY/kWh on the test dataset [34], meaning on average the prediction was about ¥1.94 off per kWh from the actual price. MAE is insensitive to outlier magnitude (each error contributes linearly), which can be useful given occasional price spikes.

- **Root Mean Square Error (RMSE):** the square root of the average squared error. RMSE penalizes larger errors more than MAE. It's often reported alongside MAE. In the same study, RMSE was ~2.630 JPY/kWh [34]. The gap between RMSE and MAE indicates the presence of some larger error instances (spikes) since RMSE > MAE. Lower RMSE indicates better handling of volatility. Many papers in Japan and elsewhere use RMSE as a primary metric, especially when comparing to earlier works (since it's mathematically convenient for optimization and has been a traditional choice in forecasting competitions).

- **Mean Absolute Percentage Error (MAPE):** the mean absolute error expressed as a percentage of actual value. MAPE provides a scale-independent measure of error (useful for comparing performance across regions or time frames with different price levels). However, MAPE has issues when actual prices are near zero (which can happen during solar oversupply – a challenge in Japan). Despite that, it's still commonly seen in evaluation. Industry practitioners like to use MAPE because it directly tells you "on average, the forecast was X% off." It's been noted that even a 1% improvement in MAPE can translate to significant cost savings for market participants – one analysis found *each 1% MAPE improvement yielded ~0.1–0.35% cost reduction in procurement* [61]. This underscores why many utilities aim to drive MAPE down with better models.

- **Mean Squared Error (MSE) and R^2:** Some publications include MSE (the square of RMSE) or the **coefficient of determination (R^2)** as supplementary metrics. R^2 indicates the fraction of variance explained by the model. For example, the ensemble CNN–LSTM in Kyushu achieved an R^2 of about 0.787 on the test set [62] – meaning ~78.7% of the variance in prices was explained by the predictions. R^2 is less common in price forecasting papers than MAE/RMSE, but when models are compared against a naïve baseline, an R^2 (or an improvement percentage) can be illustrative.

- **Mean Directional Accuracy:** In trading contexts, one might check how often the model predicts the price *direction* correctly (up or down relative to previous interval). This is sometimes reported as a percentage. It's not as prominent in academic literature for day-ahead prices, but it can be of practical interest to traders. We mention it for completeness, though we haven't seen it emphasized in Japan-specific studies reviewed here.

- **Probabilistic Forecast Metrics:** As focus grows on **probabilistic forecasting** (providing a distribution or interval for prices rather than a point estimate), metrics like the **Pinball Loss** (for quantile forecasts) and **Continuous Ranked Probability Score (CRPS)** become relevant. For example, a recent work on probabilistic price forecasting (not Japan-specific) used pinball loss to

evaluate quantile regression models [63]. In Japan's context, if a model predicts a range of possible prices (perhaps to account for uncertainty in solar output), these metrics would measure how well the predicted probability distribution matches reality. However, most of the Japan-focused studies to date emphasize point forecasts, so probabilistic metrics are not yet common in those reports.

- **Specialized Metrics for Classification or Rare Events:** In the case of modeling spike occurrence (as a classification or point process), different metrics are used. Adline and Ikeda (2023) evaluated their spike prediction model with the **Matthews Correlation Coefficient (MCC)**, which is a balanced measure for binary classification, and a **weighted accuracy index** tailored to the sparse nature of spike events [16]. These metrics are useful when the forecast target is not a continuous price value but an event probability (e.g. "spike or no spike"). They complement using standard MAE on the predicted probabilities of spikes [64]. Another example: some studies might report precision/recall for spike prediction, or use **area under ROC curve (AUC)** if treating it as a binary classification. For the majority of price forecasting (which outputs a numeric price), such classification metrics don't apply, but they appear in niche sub-problems like predicting negative-price events or price spike days.

In practice, papers often report multiple metrics to give a full picture of performance. For instance, a model might have very low MAE but if it occasionally misses a huge spike, the RMSE will capture that (showing higher error). By examining both, one can judge consistency. In Japan's volatile market, it's not unusual to see a method tuned to minimize MAE (to be consistently accurate in normal conditions) while additional analysis is done on how it performs in extreme cases (perhaps via separate metrics or case studies on specific days). Increasingly, statistical tests are also advocated: e.g. **Diebold-Mariano tests** to check if error differences between models are significant, ensuring one model's improvement over another is not just by chance [65] [66]. Summing up, **MAE, RMSE, and MAPE** are the go-to metrics for point forecasts in the Japanese power market, often supplemented by R^2 or error reduction percentages, and more specialized measures when addressing classification-type forecasts or probabilistic outputs.

Validation Approaches and Rolling Forecasts

Evaluating forecast models on historical data must be done carefully to avoid **overestimating performance**. In time-series problems like electricity prices, it's critical to respect the chronological order of data when training and testing models. Thus, researchers commonly use **rolling forecasting** methodologies or similar time-series validation techniques:

- **Rolling Window (Walk-Forward) Evaluation:** In a rolling forecast setup, a model is trained on a certain initial period, then tested on the next time step (e.g. next day). Then the window "rolls" forward – the new actual data is added to the training set (and possibly the oldest data dropped, if using a fixed-size window) – and the model is retrained to predict the following step, and so on. This process simulates how forecasting would work in real time, always using only past data to predict future. It prevents **look-ahead bias** (where a model could otherwise accidentally be influenced by future data). **Most recent studies implement some form of rolling evaluation**, as it is considered best practice in EPF [67]. In fact, short-term electricity price models are *"typically estimated via rolling windows, using only the most recent observations"* [67].

In Japan-centric research, we see this approach used explicitly. For example, Wang *et al.* (2024) employed a *day-by-day retraining*: they incrementally extended the training dataset each day to include new information,

and generated forecasts for the next day in a continual loop [36]. Their results in Table 3 showed that this **day-by-day updated model** yielded better accuracy than a single static model would. Fang *et al.* (2024) also compared different training regimes and found that a **continuous training loop (rolling retrain)** led to superior accuracy for both ANN and SVR models [21]. These findings reinforce that re-fitting the model regularly to the latest data (especially in a changing environment with growing renewable influence) is beneficial.

- **Expanding vs. Sliding Window:** There are two variants of rolling windows – *expanding*, where you keep adding new data without removing old data (the training period grows over time), and *sliding/fixed window*, where you keep the window size constant (dropping the oldest data as new data comes in). An expanding window uses all past data and can improve statistical power, but if the process changes over time (non-stationarity), very old data might mislead the model. A sliding window limits training to the most recent N days/weeks, which can adapt to regime changes (e.g. post-2016 market liberalization or post-2020 solar boom). Some studies experiment with window length: e.g. "estimation sample length" analysis has been done to determine how much history is optimal [68]. While specifics on window size tuning in Japan were not prominent in the sources, it's generally recognized that including at least one year (to capture seasonal cycle) is useful, but not much more than a few years if the market undergone structural changes (like new policies or capacity changes). In any case, the **rolling evaluation ensures each test prediction is truly out-of-sample** and that the model is always up-to-date.

- **Cross-Validation for Time Series:** Traditional k-fold cross-validation (randomly shuffling data) is not applicable to time series because of temporal order. Instead, forecasters use **time-based cross-validation**. One method is to define multiple train/test splits in chronological order. For example, train on 2017-2019, test on 2020; then train on 2017-2020, test on 2021, etc. This is effectively an expanding window evaluation with yearly steps. It provides multiple evaluations (one for each fold) to gauge stability. A variant is **rolling-origin evaluation** where the forecast origin (start of test) moves forward in increments. This approach was advocated by Weron (2014) in his review, calling for robust procedures that test models on the same rolling splits to enable fair comparisons [69]. In practice, many academic papers report performance on a single contiguous test period (e.g. "we trained on Jan–Dec 2022, and tested on Jan–Mar 2023" as in the MDPI study [34]). However, they often also perform a rolling test across a longer horizon internally or do **ex post analysis** on whether errors increase over time, etc. The best studies ensure that their model selection and hyperparameter tuning are themselves done on a validation set prior to final testing, to avoid overfitting to the test set.

- **Handling Concept Drift:** The Japanese power market saw a **significant shift in 2020-2021** (with the extreme price spike event in Jan 2021 reaching ¥200+ [70]). Models trained on pre-2020 data would have struggled to predict such an event. Rolling retraining helps mitigate this by allowing the model to gradually learn new patterns (e.g. that prices can reach very high levels under certain conditions). Some studies explicitly mention updating model parameters to account for new regimes (for instance, by giving more weight to recent errors or using adaptive learning rates). While not always detailed, in industry practice it's common to monitor model error over time and trigger retraining or model revision if performance degrades (a sign of concept drift).

- **Benchmarking and Robustness Checks:** Aside from rolling evaluation, papers often compare their model against simple benchmarks like a **naïve forecast** (e.g. "tomorrow's price = today's price") or a

simple average model. This provides context for how much value the complex model is adding. Additionally, statistical tests (Diebold-Mariano for predictive accuracy, etc.) are used to confirm that differences in error metrics are statistically significant over the test sample [71]. The 2014 review by Weron emphasized using **the same datasets and robust error evaluation procedures** across models, implying rolling tests and significance testing, to truly judge if one method outperforms another [69].

In the reviewed studies, it's clear that **rolling forecasting methodology is considered a gold standard for model validation in electricity price forecasting**. By retraining models as new data arrives and always testing on unseen future data, researchers ensure that their reported accuracy is realistic. For Japan in particular, where market conditions have evolved quickly with policy changes and renewable integration, this approach is indispensable. The adoption of rolling validation is evidenced by phrases like *"training data are incremented daily to include the most recent information"* [36] and *"continuous loop method performed better"* [21] in the literature. This mirrors real-world operations, where a utility or trader would re-fit their forecast model frequently (even daily) as fresh market data comes in. It also allows methods to adapt to seasonal variations (e.g. different error patterns in summer vs winter).

In conclusion, careful validation via rolling forecasts and time-aware splits gives confidence that the models discussed (CNN–LSTM, Lasso, SVR, etc.) will generalize to future price data and not just past patterns. As the Japanese power market continues to change (e.g. possible future negative prices or new bidding behaviors), such adaptive modeling and validation techniques will remain essential to maintain forecasting accuracy.

Conclusion

Power price forecasting in Japan has advanced significantly, blending foundational time-series techniques with cutting-edge machine learning. Traditional models like ARIMA and GARCH laid the groundwork but proved insufficient alone in the face of renewable-driven volatility and price spikes. In the past five years, there's been a clear shift toward **data-rich, AI-driven models** – with LSTM neural networks, hybrid CNN-LSTM frameworks, and ensemble tree methods demonstrating superior performance in capturing the complex dynamics of JEPX prices [17] [25]. These models have been enriched by a diverse set of features: from conventional ones such as demand, weather, and past prices, to innovative additions like net demand, GIS-based solar metrics, and even web search trends, all aiming to encompass the multifaceted factors influencing prices [72] [49]. Feature engineering (rolling statistics, decompositions, etc.) and rigorous validation (rolling windows and continuous retraining) further ensure that these models remain robust and applicable to real-world operations [43] [36].

It is also evident that **no one model fits all situations** – success comes from tailoring the approach to the Japan market's specifics. For example, accounting for solar PV's impact (which is particularly pronounced in Japan's midday price shape) via net demand or meteorological inputs is now considered essential [38]. Similarly, recognizing the importance of calendar effects (e.g. midday price dips on weekends or holidays) can greatly enhance accuracy [40]. Studies that focused on these aspects achieved more reliable forecasts and provided insights that help market participants (like PV plant operators or retailers) make informed decisions [19] [40].

Looking forward, as Japan's market continues integrating renewables and perhaps moves toward more **deregulated and transparent trading** (e.g. growth of an intraday market or aggregated flexibility markets), forecasting methods will likely evolve further. We can expect greater emphasis on **probabilistic**

forecasts (to quantify uncertainty in prices), use of **transfer learning** from other markets, and perhaps real-time data sources (like live demand-generation feeds or advanced weather nowcasts). Techniques like Transformers or reinforcement learning agents for bidding could make their way into practice, building on the strong foundation highlighted in current research.

In summary, the state-of-the-art for power price forecasting in Japan is a synergy of domain knowledge and modern AI: it leverages the known drivers of price (demand-supply fundamentals, weather, calendar) within sophisticated models validated by robust techniques. This combined approach has yielded impressive accuracy (often with MAE errors on the order of just a few JPY) [34], and it equips stakeholders in Japan's electricity market with valuable predictions to navigate the uncertainties of a cleaner but more volatile energy future.

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